

A randomized, 48-week, placebo-controlled trial of intensive lifestyle modification and/or metformin therapy in overweight women with polycystic ovary syndrome: a pilot study

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Received September 30, 2003; revised and accepted February 4, 2004.

Supported by a Mae Stone Goode foundation grant (KMH); Women's Reproductive Health Research Career Center grant K12 HD01332-03 to the University of Rochester; and General Clinical Research Center grant 5 M01 RR00044, National Center for Research Resources, National Institutes of Health.

Presented in part at the 58th Annual Meeting of the American Society for Reproductive Medicine, Seattle, Washington, October 12–17, 2002.

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0015-0282/04/\$30.00
doi:10.1016/j.fertnstert.2004.02.104

Objective: To obtain data from a pilot randomized trial on the effect of metformin therapy and lifestyle modification on ovulation and androgen concentrations in women with polycystic ovary syndrome (PCOS).

Design: Prospective, randomized, placebo-controlled pilot trial.

Setting: Academic medical center.

Patient(s): Thirty-eight overweight or obese women with PCOS.

Intervention(s): All subjects were randomized to one of four 48-week interventions: metformin 850 mg two times per day, lifestyle modification plus metformin 850 mg two times per day, lifestyle modification plus placebo, or placebo alone.

Main Outcome Measure(s): Recruitment, dropout, and compliance with a long-term lifestyle intervention in PCOS; preliminary estimates of treatment effect on ovulation, as measured by weekly urinary pregnanediol glucuronide, and on total T and free androgen index.

Result(s): It was necessary to screen seven women to have one subject randomized. The dropout rate was 39%, with the majority of dropouts occurring within the first 24 weeks. Mean body mass index was >39 mg/kg². Modest weight reduction was found in all treatment groups, with the most significant reduction occurring with the combination of metformin and lifestyle intervention. Significant androgen reduction occurred in the combination group only. Ovulation rates did not differ significantly between groups. However, when data were analyzed by presence or absence of weight reduction in subjects, independent of treatment group, the estimated odds ratio for weight loss was 9.0 (95% confidence interval 1.2–64.7) with respect to regular ovulation. If weight loss occurred during metformin therapy, the odds ratio for regular ovulation was 16.2 (95% confidence interval 4.4–60.2).

Conclusion(s): Key methodologic issues for a large-scale, randomized trial of lifestyle intervention in PCOS include minimizing early dropout from the lifestyle intervention and including a range of body mass index that is not skewed toward severe obesity. Weight reduction might play the most significant role in restoration of ovulation in obese women with PCOS. (Fertil Steril® 2004;82:421–9. ©2004 by American Society for Reproductive Medicine.)

Key Words: Polycystic ovary syndrome, obesity, ovulation

The polycystic ovary syndrome (PCOS) is a common disorder of reproductive-aged women, affecting 5%–10% of such women (1). It is characterized by hyperandrogenism and oligo- or anovulation (2). Women with PCOS are often overweight, and obesity seems to worsen the endocrinologic disturbances of the disorder (3). Although the exact pathophysiologic de-

fect of PCOS remains unknown, it is characterized by insulin resistance accompanied by compensatory hyperinsulinemia (4). In turn, there is evidence that hyperinsulinemia plays a causal role in the hyperandrogenism of the syndrome (5–9).

Several studies have demonstrated reduction in hyperandrogenism and resumption of

ovulation with the use of insulin-sensitizing agents to reduce hyperinsulinemia in women with PCOS (10–16). The most well studied insulin-sensitizing agent is metformin, which is generally associated with improved androgen levels and resumption of ovulatory cycles or improvement in menstrual cyclicity (10–13). However, in one study in which very obese women with PCOS were treated with metformin but instructed in weight maintenance, there was no reduction in androgens (17).

Obese women with PCOS, and in particular those with abdominal obesity, seem to have adverse metabolic and cardiovascular profiles (18). Severe caloric restriction with weight reduction has been associated with resumption of menses and reduction in total and free T (19). Severely restrictive diets are notable for rapid weight reduction, but long-term compliance with weight reduction is poor, with a significant percentage of subjects returning to pretreatment weight within 1 year (20). Modest weight reduction with lifestyle modification has also been associated with restoration of ovulation and fertility in overweight women with PCOS (21). This approach might lead to more enduring weight loss than severe caloric restriction, but randomized data are not available for more than a 6-month intervention.

A combined approach of weight loss and medication was reported in only one previous study (22). There was a significant reduction in T with the use of combined weight loss and medication but not with weight loss alone. This study followed women over 6 months and did not measure ovulation directly.

In summary, obese women with PCOS are oligo-ovulatory and have an adverse metabolic and cardiovascular profile. The most effective long-term treatment strategy for these women is unknown; although metformin therapy and weight reduction have both been shown to enhance ovulation and improve the metabolic profile in short-term (6 month) studies, there are no data from a head-to-head comparison over a longer term of treatment to guide decisions in the choice of treatment.

In view of these considerations, we carried out a pilot study to evaluate two treatments, metformin therapy or lifestyle modification, as well as the combination of the two, over a longer (i.e., 12-month) period. Specifically, we wished to examine the impact of these treatments, as compared with placebo, on the return of ovulation and the reduction in serum androgens. The specific aims were to obtain the overall dropout rate from each arm, evaluate the success of a modest weight-reduction regimen modified from the Diabetes Prevention Trial (23) over the long-term, and obtain preliminary group differences on outcome measures. Although the study was not powered to produce significant differences in outcomes, the lack of long-term compliance and drop-out data available from other weight reduction studies in PCOS motivated the need for these pilot data to assess the feasibility of a larger study.

MATERIALS AND METHODS

Subjects

Women were recruited for the study by direct advertisement, referral from physician offices, and as outpatients attending the reproductive endocrinology clinic at the University of Rochester. Polycystic ovary syndrome was diagnosed by fewer than six menses per year and evidence of hyperandrogenism. All subjects were overweight or obese, with a minimum body mass index (BMI) of $>25 \text{ mg/kg}^2$. Initial evaluation was performed by telephone interview, and eligible subjects presented to the clinic for hormonal evaluation. Two hundred seventy-three women were screened by telephone interview, of whom 49 met historical criteria and agreed to hormonal screening.

Normal TSH, prolactin, FSH, and metabolic profile were required for eligibility. Adrenal disease was evaluated according to DHEAS, serum 17-OH-progesterone, and 24-hour urinary free cortisol values. Values in the normal range for the clinical laboratory were required for eligibility. Thirty-eight women met criteria for randomization and agreed to randomization. All subjects had a serum total T level $>50 \text{ ng/dL}$ in our laboratory. No subjects had taken any hormonal medication within the last 2 months before entry into the study, and no subjects were actively dieting at time of entry.

The protocol was approved by the Human Subjects Review Board at the University of Rochester, and all subjects gave their informed consent before enrolling in the study.

Study Protocol

After confirmation of eligibility, women with PCOS were studied at the General Clinical Research Center (GCRC) at the University of Rochester. All women were following their usual diet, provided that at least 250 g–330 g of carbohydrates were ingested in the 3 days preceding investigation. Blood samples for hormonal evaluation were drawn after an overnight fast between 8 AM and 9 AM. Negative serum hCG was obtained on all subjects at each GCRC visit. An oral glucose tolerance test was administered with 75 g of oral glucose at time 0. Samples were then taken from an indwelling intravenous catheter at 30, 60, 120, and 180 minutes for glucose and insulin determination.

Bioelectrical impedance was performed at the baseline visit for percent body fat determination, calculated by the following formula: percent body fat = $0.556 \times (\text{height}^2/\text{resistance} + 0.096 \times \text{weight}) + 1.73$ (BIA-101A Bioelectrical Impedance Analyzer; RJL Systems, Detroit, Michigan). At 24- and 48-week visits, bioelectrical impedance and dual energy x-ray absorptiometry (DEXA-QDR 4500 Elite; Hologic, Bedford, Massachusetts) were used to estimate total body fat at the GCRC. A general physical examination was performed.

Height was measured to the nearest 0.5 cm, and weight in kilograms was recorded. Waist and hip circumference was

obtained according to established criteria (24). A transvaginal ultrasound (Siemens, Ehrlagen, Germany) was performed to assess ovarian morphology and to obtain estimates of uterine dimensions, volume, and endometrial thickness. The ultrasound confirmed PCOS morphology but was not used in diagnosis of PCOS. An assessment of hirsutism was performed according to a modified Ferriman-Gallwey scoring system (25).

After assessment, all women were randomly assigned to one of four groups: metformin 850 mg by mouth two times per day (9 subjects), a lifestyle modification program with placebo by mouth two times per day (11 subjects), lifestyle modification with metformin 850 mg by mouth two times per day (9 subjects), or placebo by mouth two times per day alone (9 subjects). The randomization schedule was computer generated in blocks by an independent pharmacy representative, and the block schedule was blinded to the investigators. Metformin hydrochloride was obtained as a powder (Spectrum Chemical Manufacturing, New Brunswick, New Jersey) and formulated with the appropriate dose into capsules by the Investigation Pharmacy Service at the University of Rochester. Identically appearing placebo capsules were also formulated. Drug and placebo were packaged and labeled according to subject number by the pharmacy in a double-blind fashion.

All subjects were given a 1-month supply of capsules and returned for monthly visits, at which time the remaining capsules were counted and a new supply of capsules given. Subjects assigned to the lifestyle modification arms of the study received additional instruction at the time of randomization (see Lifestyle Modification Program).

Subjects returned at monthly intervals to evaluate compliance with medications and to assess side effects. Weekly morning urine samples were collected by the subjects at home and stored in a freezer. These samples were brought in monthly for storage at -20°C . At 24 and 48 weeks, subjects returned to the GCRC for identical assessment as described previously. All subjects were asked to keep menstrual diaries, and these diaries reviewed at the monthly visit. Subjects who had not experienced menstrual bleeding for a >3 -month interval were evaluated with serum hCG and progesterone and administered oral medroxyprogesterone acetate, 10 mg for 7 days, to induce menstrual bleeding.

Seven subjects (3 from the metformin-alone group, 2 from lifestyle plus metformin, 1 from placebo alone, and 1 from lifestyle plus placebo) dropped out in the first 4 weeks of the study. Of the 5 subjects dropping out who were taking metformin, 2 cited adverse effects as a contributing reason. Time commitment was cited by 3 subjects (1 placebo and 2 lifestyle plus placebo) as the main reason for withdrawal. Two subjects were unable to be contacted. Five subjects dropped out between weeks 8 and 16 (2 lifestyle plus placebo, 2 lifestyle plus metformin, and 1 placebo). Three cited time commitments, and 2 were unable to be contacted,

leaving 26 subjects who could be evaluated at 24 weeks. Two subjects (both in the lifestyle plus placebo arm) became pregnant in weeks 32 and 44, respectively. An additional subject (metformin group) was lost to follow-up after week 32, leaving 23 subjects to be evaluated at week 48. Figure 1 demonstrates the flow of study participants.

Lifestyle Modification Program

The lifestyle modification program was a comprehensive approach in the areas of nutrition, behavior, and physical activity. The lifestyle modification team consisted of a registered dietitian and exercise physiologist. Subjects were provided with an individualized healthy balanced meal plan with regular food, aiming for a 500–1000 calorie deficit per day.

Average weight reduction goals were 7%–10%. Individualized exercise programs were recommended to achieve 150 minutes of exercise per week. The nutritional composition of the meals was 50% of calories from carbohydrate, 25% from protein, and 25% from fat. Intake of low glycemic index foods was encouraged. The program's intensive phase was 24 weeks in duration, with weekly interactive group education meetings and progress monitoring. Two groups were started during the study, with 10 subjects per group. The maintenance phase was 24 weeks in duration, consisting of biweekly progress monitoring and group support meetings.

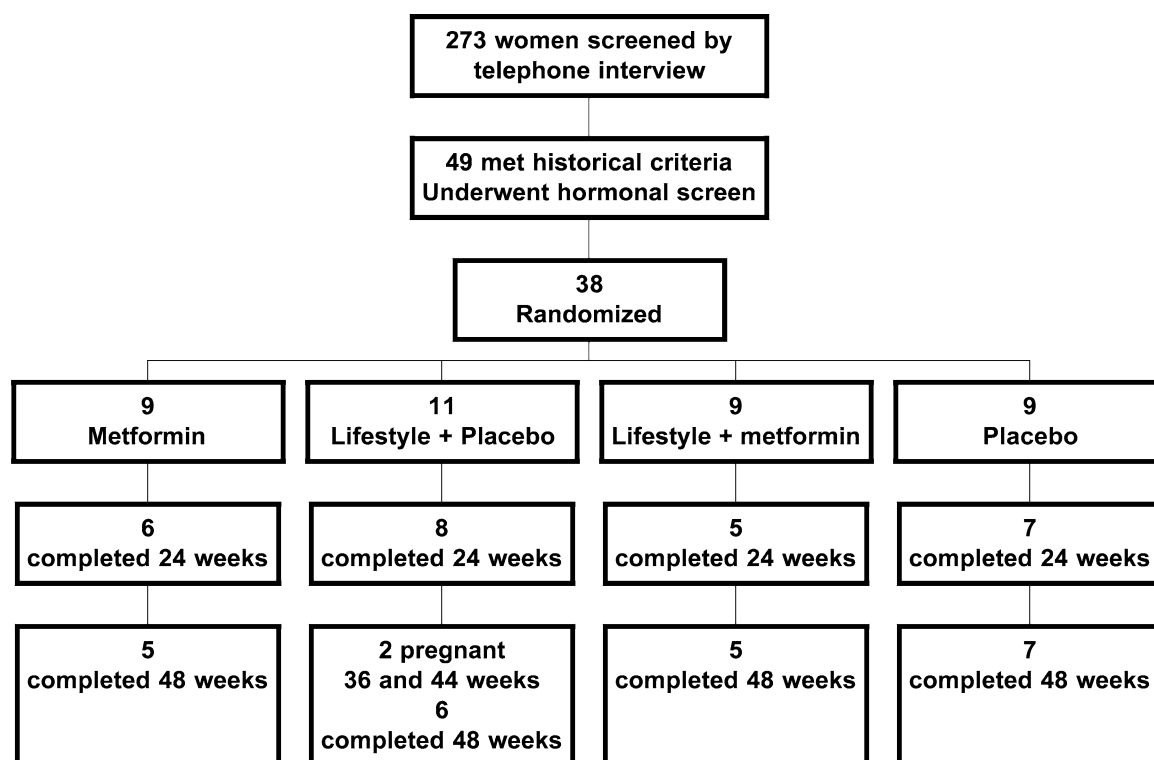
Assays

Glucose determination was made by a commercial analyzer (2700 Select Biochemistry Analyzer; Yellow Springs Instruments, Yellow Springs, Ohio). Insulin and total T determinations were by RIA (DPC, Los Angeles, California). Sex hormone-binding globulin (SHBG) determination was performed by immunometric assay (Immulite; DPC). Intra- and interassay coefficients of variation were 3% and 7% for glucose, 3%–6% and 6%–12% for insulin and T, and 6.1% and 8.0% for SHBG. Free androgen index was calculated as $\text{total T (nmol/L)}/\text{SHBG (nmol/L)} \times 100$.

Urinary pregnanediol glucuronide assay was modified from a previously reported assay (26). The purified monoclonal antibody, Quidel anti-pregnanediol-3-glucuronide, clone 330, (University of California-Davis, Davis, California) was diluted in coating buffer (0.05 mol/L Na_2CO_3 and NaHCO_3 , pH 9.6), coated onto 96-well plates (Nunc-Immuno MaxiSorp Surface; Fisher Scientific, Pittsburgh, Pennsylvania), 50 μL per well, and incubated at 2° – 8°C for 4 days. The plates were then washed with a solution of 1.5 mol/L NaCl, 0.5% Tween 20, and 20 μL enzyme immunoassay (EIA) buffer (0.2 mol/L sodium phosphate buffer, pH 7.0 with 8.7 g NaCl and 1 g bovine serum albumin per liter) was added to each well. The plates were incubated for .50–2.0 hours at room temperature and then prediluted standards, controls, or specimens were added, 50 μL per well, followed by pregnanediol-3-glucuronide horseradish-peroxidase conjugate, (diluted 1:40,000 in EIA buffer), 50 μL per well. The plates were shaken on a plate shaker (700 rpm) for 2 minutes and incubated at 2° – 8°C overnight.

FIGURE 1

Flow diagram of participants.



Hoeger. Lifestyle modification and metformin in PCOS. *Fertil Steril* 2004.

The plates were then brought to room temperature, washed, and 100 μ L of a citrate buffer, pH 4.0 with 2,2 azino-bis (3-ethybenzthiazol-6-sulfonic acid) (Sigma, St. Louis, Missouri) and 2% H₂O₂, was added to each reaction. After a 20–25-minute incubation at room temperature on the plate shaker (300 rpm), the reaction was stopped by adding 100 μ L per well of the stop solution (28.8 mol/L hydrofluoric acid and 1.0 mol/L NaOH). Optical density measures were made with a microtiter kinetic analyzer with the test and reference wavelengths, 405 and 560 nm, respectively. Pregnanediol glucuronide concentrations were obtained from optical density with a four-parameter curve and converted to μ g/mL.

Urine controls were obtained in-house and used in all assays. All controls and specimens were diluted with EIA buffer. An 8-point standard curve (1–5000 pg per well) was prepared with a commercial standard (5 β -pregnane-3 α -20 α -diol glucuronide; Sigma) that was diluted in EIA buffer.

Statistical Analysis

Results are reported as mean values $\pm$ SD unless otherwise stated. The response of glucose and insulin to the glucose tolerance test was analyzed by calculating the area under the curve (AUC) by the trapezoidal method. Baseline

group differences were analyzed by analysis of variance. Baseline group differences between completed subjects and controls were compared by independent *t*-test. Within-group differences were analyzed as difference from baseline separately at 0–24 and 0–48 weeks by paired *t*-test. Mean change scores for treatment groups from baseline to 24 and baseline to 48 weeks were compared with the mean change score in the placebo group by independent *t*-test.

Logistic regression was used to evaluate the effect of weight reduction independent of group assignment. Statistical evaluations were performed with commercial software (SPSS, Chicago, Illinois) on a personal computer. Two-tailed *P* values $<.05$ were used to define statistical significance.

RESULTS

Baseline characteristics of all subjects are shown in Table 1. Characteristics for subjects completing the study are reported in Table 2, along with the baseline characteristics of those dropping out of the study.

Clinical and Metabolic Parameters

As shown in Table 3, significant weight reduction occurred at 48 weeks in all treatment groups but not in the

TABLE 1

Baseline characteristics.

	Metformin (n = 9)	Lifestyle (n = 11)	Lifestyle + metformin (n = 9)	Placebo (n = 9)	P
Age (y)	29.5 ± 6.4	27.1 ± 4.3	30.4 ± 5.4	27.1 ± 4.5	NS
BMI	37.1 ± 4.9	40 ± 7.4	41.7 ± 6.2	37.1 ± 4.6	NS
Parity	0.7 ± 1.0	0.3 ± 0.5	1.0 ± 1.3	0.8 ± 0.8	NS
WHR	0.96 ± 0.07	0.9 ± 0.06	0.89 ± 0.07	0.94 ± 0.09	NS
Waist circumference (cm)	116.8 ± 14.6	110.4 ± 14.5	116.9 ± 15.2	112.9 ± 11.3	NS
Total T (ng/dL)	61.2 ± 23.8	56.9 ± 18.9	70.0 ± 17.1	58.1 ± 17.3	NS
SHBG (nmol/L)	22.42 ± 7.15	32.84 ± 14.96	29.91 ± 11.74	31.19 ± 10.13	NS
FAI	10.88 ± 7.47	7.14 ± 4.41	9.84 ± 5.18	6.83 ± 2.55	NS
AUC _{glu} (mg/dL/min)	25661 ± 5749	24536 ± 5592	26923 ± 3109	26411 ± 4661	NS
AUC _{ins} (μU/mL/min)	36602 ± 30938	18125 ± 9777	22899 ± 17054	26421 ± 18897	NS
FBS	101.7 ± 14.9	101.4 ± 15.8	100.5 ± 13.1	103.1 ± 15.6	NS
FI	21.59 ± 11.13	20.45 ± 9.62	24.62 ± 7.18	21.08 ± 7.43	NS

Note: All data are shown as mean ± SD. WHR = waist/hip ratio; FAI = free androgen index: T (nmol/L)/SHBG (nmol/L) × 100; AUC_{ins} = area under the curve for insulin; AUC_{glu} = area under the curve for glucose; FBS = fasting blood glucose, FI = fasting insulin.

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placebo group. Overall, the combined group had the most significant weight reduction, meeting the lifestyle goal of 7%–10% weight reduction.

Hirsutism score, as measured by the Ferriman-Gallwey score, is reported in Figure 2. Mean Ferriman-Gallwey score remained >10 for all groups after 48 weeks, and there were no significant changes in Ferriman-Gallwey score in any of the groups.

Percentile change from baseline, for both 24 and 48 weeks, in serum total T values, SHBG, and free androgen index are shown in Table 3. Significant reduction in androgens, both total T and free androgen index, is noted at 48 weeks in the combined group only. Sex hormone-binding globulin levels did not change significantly in any group.

TABLE 2

Baseline values for subjects completing and not completing the 48-week trial.

	Completed 48 weeks (n = 23)	Dropped out of study (n = 15)	P
Age (y)	29.4 ± 5.7	27.7 ± 5.1	NS
BMI	39.7 ± 7.1	37.9 ± 4.1	NS
Total T (ng/dL)	59.4 ± 18.4	60.5 ± 20.3	NS
SHBG (nmol/L)	31.6 ± 11.7	25.7 ± 11.4	NS
FAI	7.35 ± 3.23	10.7 ± 7.0	NS
AUC _{glu} (mg/dL/min)	25221 ± 2943	25891 ± 6371	NS
AUC _{ins} (μU/mL/min)	28044 ± 22993	21845 ± 16019	NS
FBS	101.2 ± 12.5	102.3 ± 17.3	NS

Note: All data are shown as mean ± SD. FAI = free androgen index (defined in Table 1 footnote); AUC_{glu} = area under the curve for glucose; AUC_{ins} = area under the curve for insulin; FBS = fasting blood glucose.

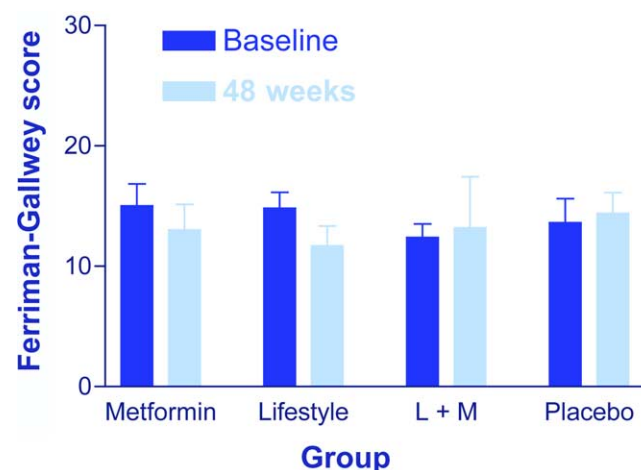
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Percent change scores for 0–24 weeks and 0–48 weeks for oral glucose stimulated values of glucose and insulin are reported in Table 3 as AUC_{glu} and AUC_{ins}. There were no significant differences for any parameter.

Fasting serum glucose concentrations showed no significant change, although all treatment groups trended toward lower fasting levels at 48 weeks. Women in the placebo group demonstrated a nonsignificant increase. At baseline, abnormal glucose tolerance, defined as a fasting blood glucose >126 ng/dL or a 2-hour glucose >200 ng/dL during an

FIGURE 2

Modified Ferriman-Gallwey hirsutism score before and after 48 weeks of treatment. L = lifestyle modification; M = metformin.



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TABLE 3

Percent change from baseline at 24 and 48 weeks.

	Weeks	M	L + PL	L + M	PL
Body weight (kg)	0–24 (n = 26)	−2.6 ± 4.6	−2.8 ± 5.6	−6.3 ± 4.4	−0.3 ± 3.3
	0–48 (n = 23)	−6.5 ± 3.7 ^a	−6.8 ± 3.8 ^a	−8.9 ± 2.9 ^{a,b}	0.2 ± 0.8
Total T (ng/dL)	0–24	−12.3 ± 22.7	−17.0 ± 31.8	−4.5 ± 24.6	7.4 ± 29.2
	0–48	6.6 ± 27.8	2.9 ± 14.1	−26.5 ± 13.9 ^{a,b}	8.3 ± 31.0
SHBG (nmol/L)	0–24	−2.5 ± 19.6	6.5 ± 36.2	0.8 ± 26.6	−0.1 ± 15.0
	0–48	2.0 ± 39.5	−14.4 ± 43.1	−3.5 ± 39.7	−26.2 ± 12.0
FAI	0–24	−4.1 ± 23.7	−37.9 ± 54.8	−15.2 ± 59.8	13.1 ± 30.9
	0–48	13.5 ± 57.1	39.6 ± 49.7	−9.0 ± 45.1 ^{a,b}	47.5 ± 32.2
AUC _{glu} (mg/dL/min)	0–24	−8.4 ± 21.6	5.2 ± 14.2	−13.9 ± 14.8	−7.1 ± 16.6
	0–48	3.5 ± 23.0	1.9 ± 10.4	−1.7 ± 19.6	2.5 ± 18.9
AUC _{ins} (μU/mL/min)	0–24	−32.7 ± 36.5	14.3 ± 12.2	−20.5 ± 39.5	−15.4 ± 34.5
	0–48	−46.0 ± 18.4	−24.2 ± 42.6	−24.4 ± 17.0	−26.3 ± 26.0
FBS	0–24	−12.5 ± 12.5	−3.8 ± 16.9	−13.5 ± 11.4	−3.9 ± 18.4
	0–48	−6.9 ± 11.3	−2.4 ± 14.9	−5.7 ± 17.0	3.1 ± 15.4

Note: Data are mean ± SD of percent change. M = metformin; L = lifestyle modification; PL = placebo; FAI = free androgen index; AUC_{glu} = area under the curve for glucose; AUC_{ins} = area under the curve for insulin.

^a *P* < .05 vs. baseline.

^b *P* < .05 vs. placebo.

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oral glucose tolerance test, was noted in two subjects in the metformin group, three subjects in the lifestyle group, no subjects in the combined group, and no subjects in the placebo group. At 48 weeks, no subject in the metformin, lifestyle, or combined groups demonstrated elevated fasting glucose >126 ng/dL or 2-hour sample >200 ng/dL, whereas two subjects in the placebo group demonstrated either fasting blood glucose >126 ng/dL or a 2-hour sample >200 ng/dL at 48 weeks.

Ovulation

Ovulation was determined by measurement of urinary pregnanediol glucuronide. If elevation in pregnanediol glucuronide was noted preceding menstrual flow, this was counted as an ovulatory event for that individual. Consecutive weekly elevated levels were counted as a single ovulatory event. The mean number of documented ovulations was 4.3 in the metformin group, 3.5 in the lifestyle group, 3.2 in the combined group, and 2.7 in the placebo group. The mean number of reported menstrual events was 7.0 for the metformin group, 6.2 for the lifestyle group, 6.6 for the combined group, and 4.7 in the placebo group. Figure 3 shows the ovulatory frequency per group broken into categories of <3, 3–6, and >6 ovulatory events. Overall, no significant differences were demonstrated in ovulatory events between treatment groups and the placebo group.

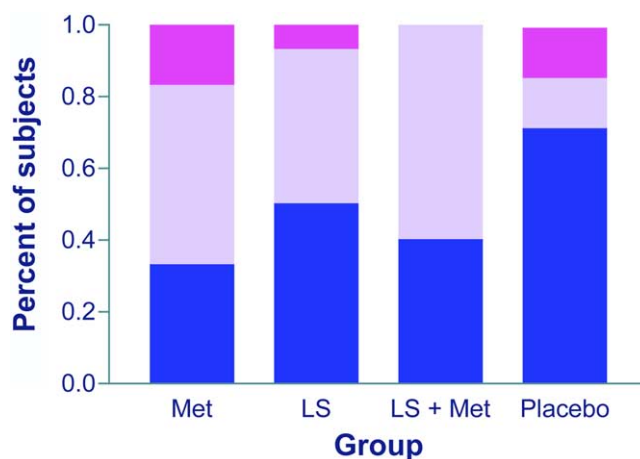
Subgroup Analysis

Huber-Buchholz et al. (21) showed that a modest reduction in weight (2%–5%) resulted in restoration of ovulation in 50% of anovulatory women. Using this literature guideline, we used a 3% cut-off point to dichotomize our sample into those who did or did not lose weight. Defined in this manner, weight reduction seemed to be the key factor af-

fecting ovulation, independent of treatment group. Figure 4 shows the mean ovulatory number over 24 weeks broken out overall by presence of weight reduction and by whether or not metformin was used. Table 4 shows the results of a logistic regression model, in which the dependent variable is whether or not the subject experienced three or more ovulations in a 24-week period. (Because of dropout after 24 weeks, the sample size at 48 weeks was too small for logistic

FIGURE 3

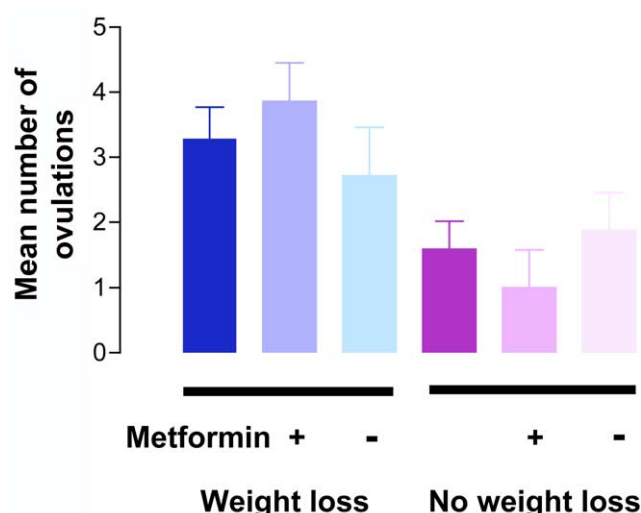
Percent of subjects in each group experiencing <3 ovulatory episodes (blue), 3–6 ovulatory episodes (light purple), and >6 ovulatory episodes (red) over 48 weeks. Met = metformin; LS = lifestyle modification.



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FIGURE 4

Mean number of ovulatory episodes over 24 weeks, displayed by subjects experiencing weight loss (*blue bars*) and not experiencing weight loss (*red bars*). Use of metformin is noted within each weight-loss group by a + or – sign.



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regression.) The predictor variables were whether or not the subject lost weight from baseline assessment and whether or not she took metformin. The interaction effect between these two variables was also included in the model.

On the basis of the estimated odds ratios (ORs), women who lost weight, with or without metformin, were estimated to be nine times more likely to become regularly ovulatory than those who did not lose weight (OR = 8.97, 95% confidence interval [CI] 1.24–64.68). Furthermore, those women in the metformin group who lost weight were estimated to be 16 times more likely to become regularly ovulatory (OR = 16.19, 95% CI 4.36–64.22). Owing to the small

TABLE 4

Prediction of three or more ovulations per 24 weeks by logistic regression (n = 26).

	β	OR (e^{β})	95% CI for OR	P
Weight loss ^a (n = 14)	2.194	8.97 ^b	1.24–64.68	.030 ^b
Metformin ^c (n = 11)	0.132	1.14	0.17–7.54	.891
Weight loss with metformin ^d (n = 7)	2.785	16.19 ^b	4.36–60.22	.034 ^b

^a Controlled for starting weight and metformin.

^b Controlled for starting weight and weight loss.

^c Controlled for starting weight.

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sample size, these OR estimates have wide 95% confidence limits, but the apparent importance of weight reduction and the interaction between weight reduction and metformin are provocative findings.

DISCUSSION

Historically, the treatment focus in women with PCOS was short-term (e.g., ovulation induction for infertility, oral contraceptives for irregular bleeding, and antiandrogens for hirsutism). In recent years, however, growing recognition that women with PCOS have characteristics that put them at risk for diabetes and cardiovascular disease has led to longer-term thinking. Thus, it is increasingly recommended that PCOS should be viewed as a chronic condition that requires a longitudinal treatment perspective (27).

Metformin has been shown in most (10–13, 28, 29) but not all (17) studies to ameliorate the adverse effects of PCOS over a 2–6-month period. Indeed, despite the absence of longitudinal data, use of metformin as a long-term treatment of PCOS has become widespread. Although metformin is not approved by the U.S. Food and Drug Administration for use with PCOS as the primary indication, its use for this condition has grown dramatically in recent years. Currently more than 2.5 million prescriptions are written for metformin each year. According to the Scott-Levin pharmaceutical database, the diagnosis “polycystic ovaries” accounted for nearly 3% of all prescriptions written for metformin in the year ending April 2003. That represented an increase of 265% over the prior year, and PCOS remains the most frequent diagnosis for metformin use after diabetes.

Weight loss in women with PCOS has been shown to improve reproductive and metabolic parameters on a short-term basis, mainly with severe calorie restriction (19, 22, 30, 31). More recently, lifestyle modification with more modest weight reduction has been studied as an alternative to severe caloric restriction, also on a short-term basis (21). Like metformin, lifestyle modification has been advocated clinically as a long-term solution without long-term data. The benefits of modest weight reduction have been reported for people at risk for cardiovascular disease and diabetes. Modest weight reduction has been demonstrated to reduce risk for diabetes in older men and women with impaired glucose tolerance (23). Additionally, modest weight reduction has been recognized as a key factor in prevention of hypertension (32), hyperlipidemia (33), and coronary heart disease (34).

If lifestyle modification were feasible and efficacious as a long-term treatment for obese women with PCOS, this would clearly present an attractive, physiologic approach that addresses the underlying pathophysiology of PCOS and that might have positive health effects beyond those related to PCOS. How feasible is it, however, to use weight loss as a long-term strategy in this population, and how would the clinical results compare with those of metformin therapy? The answers to these questions require data from a head-to-

head comparison of metformin and lifestyle modification as long-term treatments of PCOS. The absence of such data motivated the design of this study. The present study was designed as a pilot trial to guide the specific needs of a larger trial from the standpoint of recruitment, dropout, and compliance. It was hoped that expected effect sizes could also be estimated from the pilot study as a basis for determining the sample size needed for a larger trial.

One of the most difficult aspects of a trial involving participation in a weight-reduction arm is compliance and dropout. In this pilot study, overall dropout was 39%. The majority of subjects (34%) dropped out in the first 16 weeks of the study. Fifteen percent of study subjects cited adverse effects of medication as the reason for dropout, and 16% cited the time commitment as the principal reason for leaving the study (20% of those in the lifestyle modification arms). Other recent studies of dietary intervention in PCOS and non-PCOS subjects have reported similar compliance and dropout. Of 45 randomized subjects recently reported in a trial of two different dietary interventions in PCOS, 17 discontinued the trial (35). In a recently reported randomized trial of dietary therapy in a general overweight population, 41% of subjects did not complete a 12-month intervention (36).

The applicability of the current and other studies to the general obese PCOS population is limited by the difficulty in achieving compliance with true lifestyle modification that includes dietary intervention. However, in our study there was a reduction in the dropout rate as subjects continued in the study over the long term. Only three subjects left the study after 24 weeks, and two of these did so because of a pregnancy. Those subjects in the lifestyle group had supportive biweekly meetings after the initial 24-week intensive period. Weight loss continued in both lifestyle groups over the second half of the study, thus suggesting continued compliance with the lifestyle goals over the long term. Future trials should incorporate incentives that build over time to encourage long-term commitment. Additionally, because the greatest dropout occurs in the initial weeks of the lifestyle intervention, immediate connection with a lifestyle coach and, if possible, a group support network are essential to maintaining interest and motivation.

In the present study, the amount of weight loss with metformin alone approached that seen in the group that received specific lifestyle management instruction with placebo. Additionally, weight reduction seemed to be enhanced in the guided lifestyle intervention group when metformin was used in combination. Weight loss was approximately 30% greater overall in the combined group than in the lifestyle modification group alone. Perhaps the most provocative finding was that women who took metformin and lost weight were far more likely to resume regular ovulation than were those who did not lose weight, when the data were analyzed outside of randomization groups. To ensure that this result is not a statistical quirk of logistic regression, it

should be confirmed in a larger trial comparing metformin alone with metformin plus lifestyle management.

Regarding androgens, a significant reduction in total T was not seen in all the treatment groups. Because this was a pilot study and not powered for demonstrating reduction in androgens, this might reflect the small sample size. A reduction in total and free T was noted in the combined group, however, at 48 weeks. Although specific reduction in androgens has been correlated with improvements in insulin sensitivity in other studies (10, 14, 21), we did not demonstrate group differences in AUC_{ins} after oral glucose load, although a general trend toward reduction was noted across time.

Because this pilot study was not designed to look at the mechanisms of androgen reduction with the different approaches, we did not choose to use a euglycemic clamp to assess insulin dynamics. Regarding glucose metabolism, however, either improvement or no deterioration in overall glucose metabolism was noted in all the subjects completing the treatment arms, whereas a deterioration of glucose status occurred in two subjects in the placebo group. This is consistent with the results of the Diabetes Prevention Trial, which demonstrated a reduction in the rate of progression to diabetes from subjects with impaired glucose tolerance using either lifestyle modification alone or metformin therapy alone (23).

Our study population included women with a mean BMI of >39 (category III obesity). The independent effect of degree of obesity could not be estimated owing to the small number of subjects with BMI <30 in each group. Whether metformin or lifestyle modification, separately or together, would have a greater impact in women with less significant obesity could be assessed in a larger trial, perhaps stratified by weight.

With the information obtained from this pilot study, we can calculate the anticipated sample size for a fully powered trial. With a primary outcome of ovulation, to be able to show a 30% difference in ovulatory frequency over 48 weeks between treatment group and placebo, 21 subjects per arm would be needed, accounting for dropout. To see no greater than a 20% difference between treatment groups, 94 subjects would be needed per group. If the primary outcome were to be androgen reduction, a similar number of subjects would be required. To demonstrate a 25% reduction in total T, 33 subjects per group would be needed. To show no greater than a 15% difference between treatment groups, 70 subjects per group would be required. Clearly, a trial of this size would require multiple sites.

In summary, this pilot study has identified two key methodologic issues that must be addressed in any large-scale randomized trial of pharmacologic therapy vs. lifestyle modification as a treatment for PCOS in obese women: [1] the intervention must be designed to enhance compliance so that the dropout rate is held to a minimum, and [2] the range of BMI should not be skewed toward morbid obesity and should include a representative distribution of overweight and moderately obese women.

Despite the small sample size, significant dropout rate, and skewed BMI, our results are nonetheless provocative: weight loss, whether with lifestyle modification or metformin therapy, was found to be the key predictor of improved ovulation, and weight loss plus metformin might have synergistic effects. Clearly, more long-term data are needed on the effects of metformin on both reproductive and metabolic endpoints, given the popularity of its use in PCOS. This can best be accomplished with a large-scale, randomized clinical trial.

Acknowledgments: The authors thank the nurses and staff at the General Clinical Research Center at the University of Rochester for data collection, analysis, and their efforts on behalf of the participants; and the staff at the Strong weight management center for their assistance with the lifestyle modification program.

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